

PIONEER JUNIOR COLLEGE
 JC2 Preliminary Examination
 In preparation for General Certificate of Education Advanced Level
 Higher 2

CANDIDATE
NAME

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CT
GROUP

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INDEX
NUMBER

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BIOLOGY

Paper 4 Practical
 Junior College Year 2

9744/04

29 August 2017
2 h 30 min

READ THESE INSTRUCTIONS FIRST

Write your name, CT group and index number on all the work you hand in.
 Give details of the practical shift and laboratory, where appropriate, in the boxes provided.
 Write in dark blue or black pen.
 You are to use a soft pencil for any diagrams or graphs.
 Do not use staples, paper clips, glue or correction fluid.
 DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
 You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in the brackets [] at the end of each question or part question.

Shift	
Laboratory	
For Examiner's Use	
1	20
2	22
3	14
Total	55

- 1** You are required to investigate the glucose concentrations of solutions **S1** and **S2**.

Doctors use the analysis of urine to help diagnose some medical conditions. One such medical condition is diabetes which results in glucose being released in urine if the condition is untreated.

You are provided with:

- 15cm³ of 4% glucose, labelled **G1**
- 10cm³ of unknown glucose concentration representing urine, labelled **S1**
- 10cm³ of unknown glucose concentration representing urine, labelled **S2**
- 50cm³ of distilled water, labelled **W**
- 40cm³ of Benedict's solution, labelled **Benedict's solution**

Proceed as follows:

- 1** Carry out a serial dilution of the 4% glucose, **G1**, to reduce the concentration of glucose solution by half between each concentration of four successive dilutions, to give **G2**, **G3**, **G4** and **G5**. You will also need to set up a control, **C**.

You are required to make up at least 5cm³ of each concentration of glucose solution in the small glass vials provided.

Complete Table 1.1 to show how you will make the concentrations of the glucose solutions, **G2**, **G3**, **G4** and **G5**, show how you will set up the control, **C**.

Table 1.1

	G1	G2	G3	G4	G5
concentration of glucose solution / %	4.00	2.00	1.00	0.500	0.250
Label of glucose solution to be diluted		G1	G2	G3	G4
volume of glucose solution to be diluted / cm ³		5.0	5.0	5.0	5.0
volume of distilled water , W , to make the dilution / cm ³		5.0	5.0	5.0	5.0
<u>Description of the control, C:</u> (g) (Using a syringe,) transfer 5.0 cm³ of distilled water (into a small container) instead of 5.0cm³ of glucose solution;;					

- (a) correct precision for both glucose;**
- (b) correct precision for volume;**
- (c) equal volume of glucose and diluted water;**
- (d) minimal volume after transfer is 5cm³;**
- (e) correct calculation of concentration of glucose;**
- (f) correct label of glucose solution to be diluted;**

[4]

Test glucose solutions and unknowns with Benedict's solution. Excess Benedict's solution is to be added to the solutions and samples. Then heat the mixture.

You should record the time taken for first appearance of any different colour or precipitate from the blue starting colour.

The result will be compared with the time taken for first appearance of any different colour or precipitate obtained from glucose solutions **G1**, **G2**, **G3**, **G4** and **G5** with Benedict's solution.

To do this you need to use the same procedure.

- 2** State the volume of Benedict's solution and the volume of the solutions (**G1**, **G2**, **G3**, **G4** and **G5**) and the unknown samples you are testing (**S1** and **S2**).

Volume of Benedict's solution: **3.0**cm³

Volume of each glucose solution (**G1**, **G2**, **G3**, **G4** and **G5**): **2.0**cm³

Volume of sample (**S1** and **S2**) : **2.0**cm³

- (a) correct precision (1dp);
- (b) appropriate volume of Benedict's solution (Max 4cm³);
- (b) higher volume of Benedict's solution than glucose/sample solution;
- (b) same volume of glucose and sample;;

No marks awarded if:

- if 2cm³ of Benedict's solution used with 1cm³ of sample → rationale ppt or colour change might be too insignificant to be observed.
- if total volume is ridiculously high and would not be feasible in a test tube.

[2]

- 3 Set up a water-bath and, test each test-tube separately, test all concentrations of **G** (**G1**, **G2**, **G3**, **G4** and **G5**) and the samples **S1** and **S2** for the presence of glucose. Start timing when the test-tube is placed into the hot water-bath. If there is no colour change after 480 seconds, record 'more than 300' as your result.
- 4 Observe the test-tube very carefully for the first appearance of any different colour or precipitate from the blue starting colour and record the timing for this change.

- (a)(i) State one variable, other than volume, which needs to be kept constant when you do the tests. [1]
temperature;;

- (ii) Describe how you will control this variable constant.[1]

(a) monitor with thermometer / observing the presence of steam / observing the presence of bubbles to indicate that water is boiling at 100C;;

- (b) Use the space below to record all your results. [4]

Table of time taken for time taken for first appearance of any different colour or precipitate from the blue starting colour / s for different glucose concentration / %

Concentration of glucose solution / %	time taken for time taken for first appearance of any different colour or precipitate from the blue starting colour / s
0.00 (C)	More than 480
0.250	
0.500	
1.00	
2.00	
4.00	

- a) correct headings with units;;
- b) correct precision for glucose concentration → 3sig fig;
- c) correct precision of time → whole numbers;
- d) trend (shortest time taken for 1st appearance for 4% glucose solution → longest time taken for 1st appearance for 0.250% glucose solution) ;;
- e) title with units;;
(allow ECF for concentration)

- (c) Using your results from step 7, estimate the concentration of glucose in sample **S1** and **S2**.

Time taken for first appearance of any different colour or precipitate from the blue starting colour for **S1**:

Concentration of glucose in **S1**.....[1]

Time taken for first appearance of any different colour or precipitate from the blue starting colour for **S2**:

Concentration of glucose in **S2**.....[1]

[1]

S1 – 1.5% (actual concentration)

S2 – 0.1%(actual concentration)

- a) time taken for S1 should be between timing for 1-2% glucose solution; (trend)**
 - b) time taken for S2 should be longer than 0.250%; (trend)**
 - c) concentration of glucose for S1 with correct units : 1-2%; R 1.5% with and time (with units for both conc and time);;**
 - d) concentration of glucose for S2 with correct units : less than 0.250%, R 0.1% with and time (with units for both conc and time);;**
- Allow ECF**

- (d) Identify two significant sources of error in this procedure and for each, suggest how you would improve the procedure to minimize the source of error. [4]

- a1) no replicates done → results may not be reliable**
- a2) carry out at least 3 replicates**

- b1) subjectivity in determining the 1st appearance of ppt or colour;**
- b2) use colorimeter / spectrophotometer to determine colour change;**

- c1) too few known concentration of glucose;**
- c2) increase more glucose standards e.g. 0.125%, 0.0625% etc**

- d1) interval between each concentration of glucose is too wide;**
- d2) decrease the interval between each concentration e.g. instead of half dilution, use a 1:4 dilution so that concentration unknown can be more accurately gauged.**

[Total: 20]

2 During this question you will require access to

- a microscope fitted with eye piece graticule
- and slide **K2**.

K2 is a cross section of a portion of a leaf from plant that grows in full sunlight and are adapted to relatively high light intensities.

- (a) Examine the slide under a microscope and locate a suitable cross section for your plan diagram as seen in Fig. 2.1. In your view, you should be able to observe the distinct categories of different types of cells in a leaf cross section. Choose the lens that is most suitable for viewing the cross section of the leaf in the field of view.

Please avoid the mid rib region for your plan diagram seen in Fig. 2.1

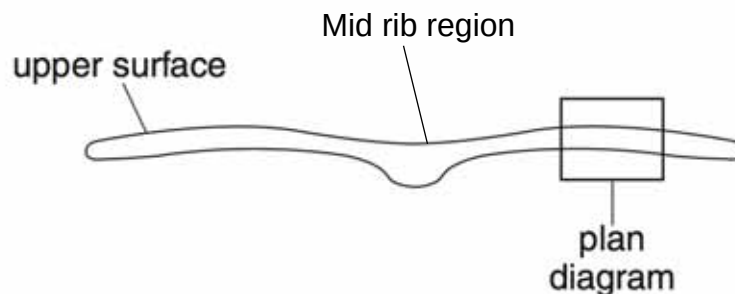


Fig. 2.1

State which objective lens you have decided to use and give a reason for your choice. [1]

- a) **40X objective lens;**
 b) **most accurately measured for total leaf thickness (idea of).**

- (b)(i) Using the objective lens selected in (a) and the eyepiece graticule fitted into your microscope, make measurements of the total leaf thickness of **K2**. [1]

No. of divisions of eyepiece graticule: **40-72 (range allowed);;**

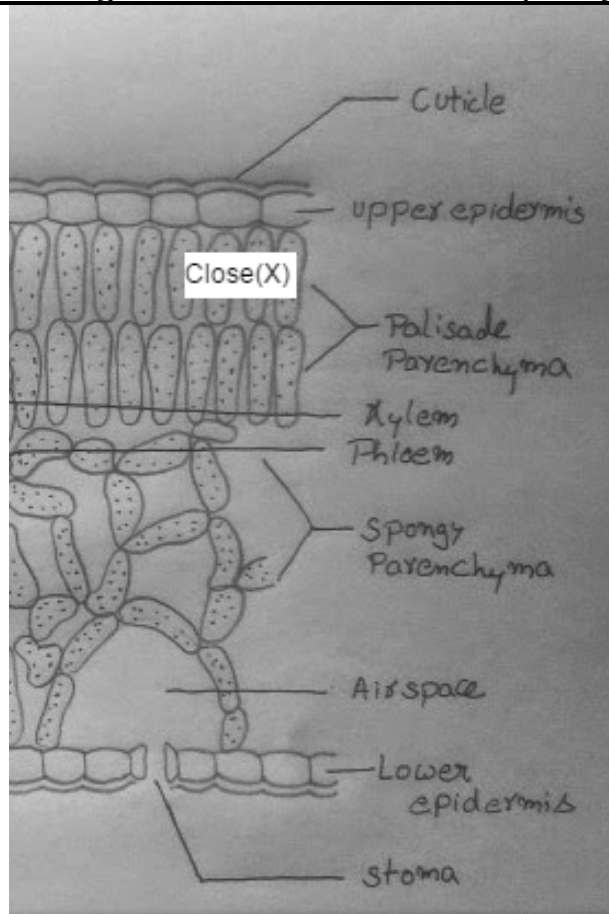
- (ii) Calculate the actual thickness of the leaf of **K2** using your data from step (b(i)). Let each division on the eyepiece graticule be 0.0025mm.

Show your working with units. [2]

- a) **correct working with units;;**
 b) **correct answers with units;; (100µm - 180µm)**
(if units missing for workings : -1)

- (c) Make a detailed, labelled drawing of a section of the cross section of **K2** in the space below. [5]

Drawing of a cross section of K2 leaf (X mag)



Marking Points

Award marks for each of the following:

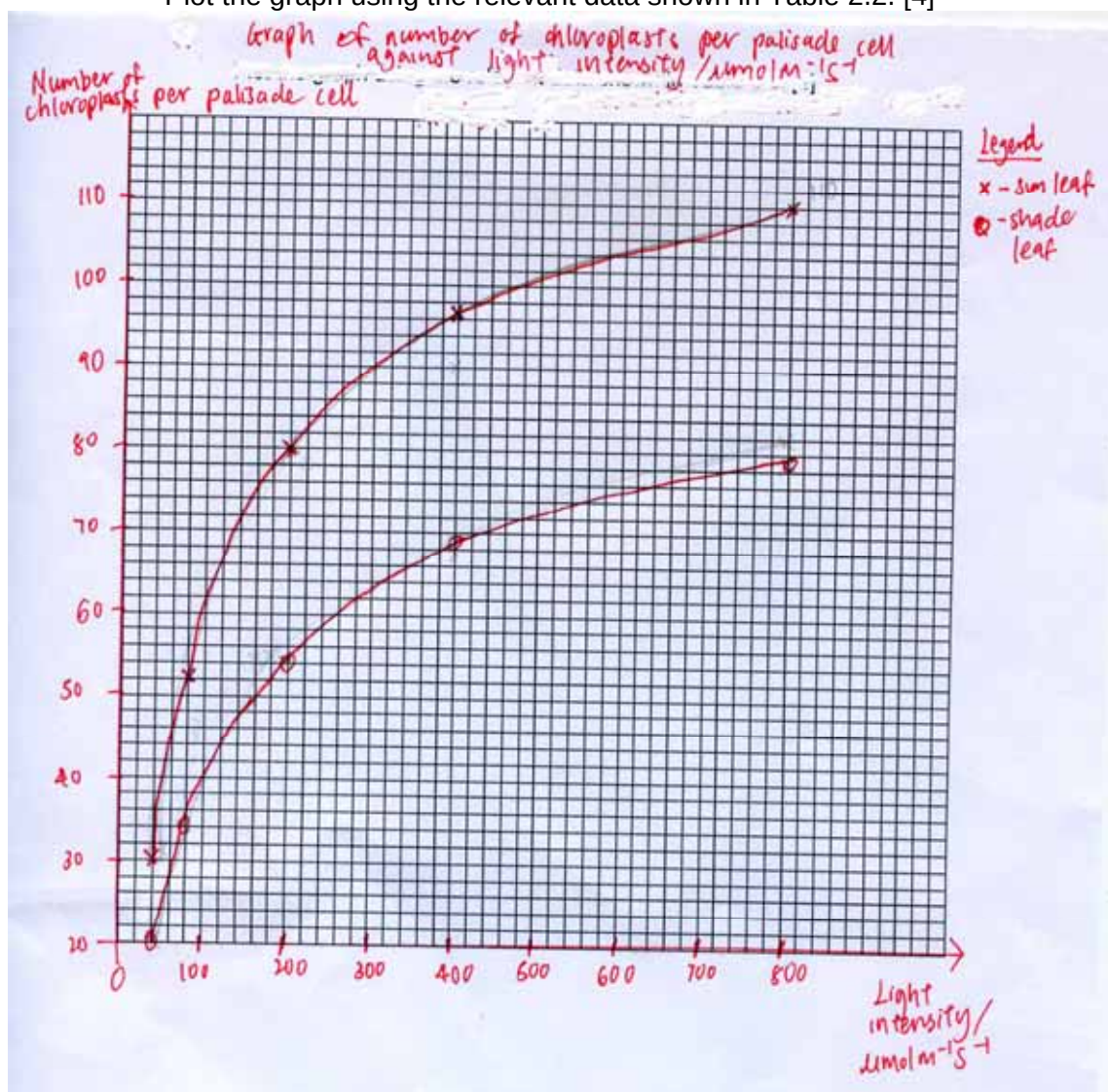
- a) appropriate title given ;
 - b) mag in title; (ECF allowed)
 - c) clear continuous lines ;
 - d) draws at least 2 layers of palisade cells;
 - e) correct shape AND proportion of spongy mesophylls and palisade cells ;
 - f) no shading ;
 - g) at least three correct labels from: spongy mesophylls / guard cell / palisade layers / airspace / cuticle / lower epidermis / upper epidermis;;
 - h) magnification calculated using total leaf thickness w units;
 - i) scale
- (d) (i) Table 2.2 shows the effect of light intensity on the number of chloroplasts between sun and shade plants.

Table 2.2

Light intensity / $\mu\text{molm}^{-2}\text{s}^{-1}$	Number of chloroplasts per palisade cell	
	Sun leaf	Shade leaf
800	110	79
400	97	69
200	80	54

80	52	34
40	30	20

Plot the graph using the relevant data shown in Table 2.2. [4]



Marking Points

Award marks for each of the following:

- appropriate title given with units ;
- appropriate size i.e. at least $\frac{3}{4}$ of graph paper ;
- correct choice of axes **AND** units ;
- intervals of the graph are equidistant **AND** no awkward scale ;
- correctly plotted point to within half a small square for both sun and shade leaf;
- appropriate line of best fit ;
- no extrapolation beyond extreme measured data ;
- use of different symbols for the 2 lines and legend is provided.

- (ii) Using the graph, describe and explain a relationship between the factors investigated. [4]

- a) As light intensity increases from 40 to 800 $\mu\text{molm}^{-2}\text{s}^{-1}$, the number of chloroplasts per palisade cell increases from 30 to 110 in sun leaf and from 20 to 79 in shade leaf;;
- b) This shows that in both sun and shade leaves; number of chloroplasts in palisade cells shows a direct relationship with light intensity, increasing in number as light intensity increases.
- c) The palisade cells of sun leaves contain more chloroplasts than those of shade leaves at each light intensity. ;
- d) This shows that the same light intensity, the rate of photosynthesis in sun plants is faster than that in shade plants.
- e) (quote any one of the light intensity and rate of photosynthesis of sun leaf and shade leaf).
- f) An increase in chloroplast numbers increases the amount of light energy absorbed which is used in cyclic and non-cyclic photophosphorylation of the light reaction to synthesise more ATP and NADPH. ;;
- g) These products of the light reaction are then used in the Calvin cycle/ light independent stages ;

to reduce carbon dioxide to produce more carbohydrates and thus increase the rate of photosynthesis. (Thus, a greater number of chloroplasts results in a higher rate of photosynthesis.); / (idea of) eventually producing more CHO / raw materials --> utilised by cell to produce more chloroplast (to capture / harness more light energy) ;

- (e) In a separate study, a student recorded the length of the stomatal density of two plants of the same species. One plant was not regularly exposed to full sunlight and often show adaptations to relatively low light intensities. The leaves of such a plant are known as shade leaves. Another plant was grown in full sunlight and are adapted to relatively high light intensities and the leaves are called sun leaves.

A statistical test was carried out to determine whether there was a significant difference in the mean stomatal density between shaded and exposed leaves.

- (i) State a statistical test that could have been used to determine whether the difference in the mean stomatal density between the shade and sun leaves is significant. [1]

t-test;;

- (ii) A summary of the student's results is shown in Table 2.3.

Table 2.3 shows the student's results.

Table 2.3

Mean stomatal density / mm ⁻²		Significance of difference	Total sample size
Shade leaves	Sun leaves		
290	335	$p < 0.05$	30

Comment on what these results show and suggest an explanation for any pattern. [4]

- a) stomatal density of sun leaves is higher than shade leaves;
- b) the difference is significant;
- c) degrees of freedom is 28;
- d) probability that difference in means would occur less than 0.05 / 5%;
- e) any 2 points:
 - i. light is likely to be the limiting factor in shade leaves/ light is not the limiting factor in sun leaves
 - ii. CO₂ concentration / availability is likely to be limiting factor in sun and not limiting factor in shade
 - iii. higher stomatal density gives higher rate of uptake of CO₂;
 - iv. allows sun leaves to make use of more availability light for photosynthesis / rate of photosynthesis

[Total: 22]

- 3 Fig. 3.1 shows an electrode connected to a data logger that can be used to measure the concentration of potassium ions in the water.



Fig. 3.1



Fig. 3.2

Design an experiment, using the electrode to investigate the effect of temperature on the permeability of potato cell membranes. Potatoes are rich in potassium ions. When small disc-shaped potatoes cut from a core borer as seen in Fig. 3.2 are placed in water, potassium ions are released from the cell into the water.

You must use:

- potatoes,

- potassium ion-selective electrode which measures in mg/L,
- a core borer of 10mm in diameter,
- distilled water.

You may select from the following apparatus and use appropriate additional apparatus:

- normal laboratory glassware .e.g boiling tubes, test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- blunt forcep,
- syringes,
- scalpel,
- ruler,
- timer e.g. stopwatch or stop clock,
- thermometer,
- isotonic buffer solution,
- hot water and ice.

Your plan should:

- have a clear and helpful structure such that the method you use is able to be repeated by anyone reading it.
- be illustrated by relevant diagram(s) to show, for example, the arrangement of apparatus used.
- identify the independent and dependent variables,
- describe the method with the scientific reasoning used to decide the method so that the results are as accurate and reliable as possible,
- show how you will record your results the proposed layout of results tables and graphs,
- use the correct technical and scientific terms,
- include reference to safety measures to minimize any risks associated with the proposed experiment.

[Total:14]

Suggested Answer

Aim (1m):

To investigate the effect of temperature[independent variable] on the permeability of cell membranes [dependent variable] using potatoes via the use of electrode and data logger to measure the concentration of potassium ions in the water [method].;;

Background (3m – including hypothesis):

a. Ions are charged;

- b. Ion channels to provide hydrophilic channel; or repelled by hydrophobic core thus cannot diffuse through the membranes
- c. Fluidity of membrane increases when temperature increases,
- d. more heat energy, phospholipids gain kinetic energy
- e. disrupts hydrophobic interactions
- f. Membrane more fluid and leaky
- g. At high temperatures, membrane disrupted / proteins denature → protein channels can be leaky as well, K^+ fully released into water
- h. Cell wall is fully permeable to ions
- i. Higher temperature, greater membrane permeability, more K^+ diffuse out

Hypothesis:

Increasing temperature would increase the permeability of the cell membranes as the weak bonds in the cell membrane are easily broken thus allowing more pores to be formed. Thus resulting in higher concentration of K^+ detected.

Experimental procedure(6m):

1. Set up the experiment in replicates of 3[a].
2. Cut a potato into cylindrical shapes of diameter 10mm using the core borer[b].
3. Cut the cylindrical potato into disc shape of height 3mm[c-consistent height] using scalpels.
4. Place the disc-shaped potatoes into an isotonic buffer[d]. This is to prevent the potato from drying up and also to maintain its osmotic pressure.
5. Add 10ml of fresh de-ionized water into a boiling tube[e] and equilibrate in a 10°C water bath for 3 mins[f] until the de-ionized water is at 10°C. Monitor temperature using thermometer.
6. Add 10 discs of potatoes[g] and start the stopwatch[h].
7. After 15min (accept range)[i], measure the concentration of K^+ with the electrode.

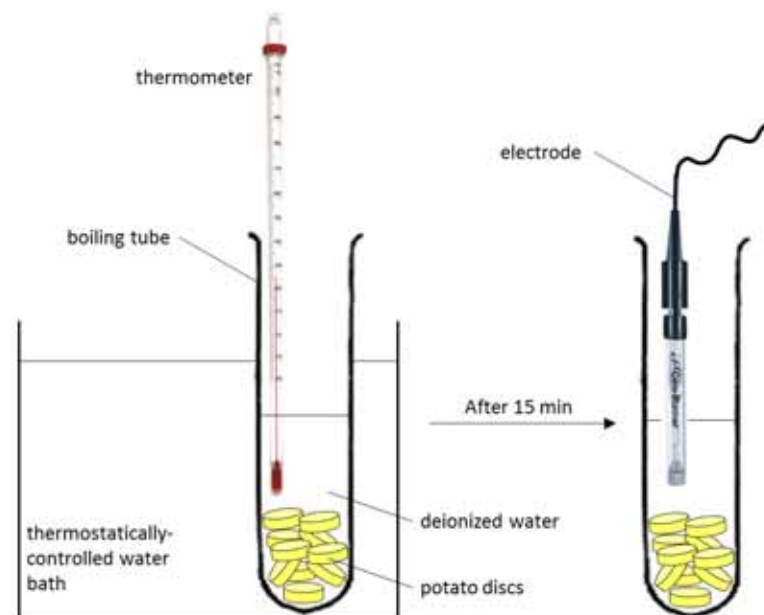


Diagram of experimental setup[j;; - with labels, correct proportion and logical setup]

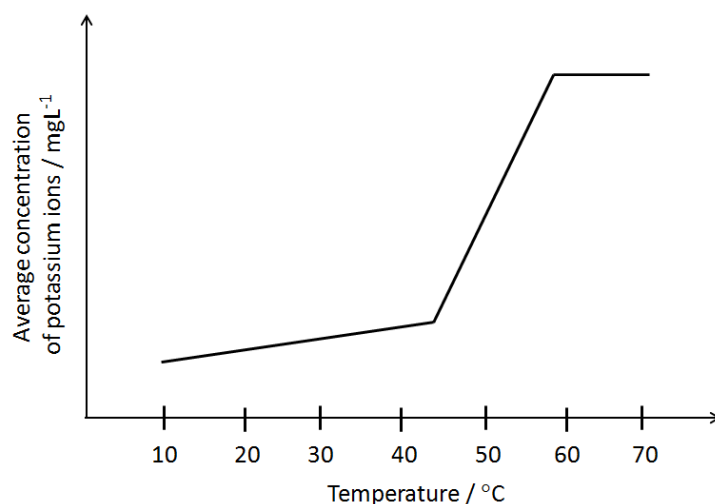
8. Repeat step 4 to 9 for 20°C, 30°C, 40°C, 50°C, 60°C and 70°C [k].

9. Record the data collected in the table below [l;;- table of recording with title, must have average and units] .

Table showing effects of temperature/°C on the concentration of K⁺ detected in the solution/mgL⁻¹

Temperature / °C	Concentration of K ⁺ / mgL ⁻¹			
	1	2	3	Average
10				
20				
30				
40				
50				
60				
70				

10. Repeat the whole experiment using fresh potatoes twice and fresh distilled water to ensure reproducibility [m] .
11. Plot a graph of average concentration of potassium ions/mgL⁻¹ against temperature (°C) [n – title and graph showing plateau, with units].



Any 7 points [3.5]

→ a to i and m

drawing – j [1]

table of recording - l [1]

graph – n [1/2]

Variables (2m)

Dependent variable : Concentration of K⁺ / mgL⁻¹ [a]

Independent variable : Temperature (10°C, 20°C, 30°C, 40°C, 50°C, 60°C, 70°C) [b- at least 5 different temperatures]

Other variables to be kept constant [c- at least any 3;;]:

- Same size and number of potato discs from the same potato

- Volume of water in all tubes
- Fresh de-ionized water in all tubes
- Time for testing
- Same electrode

Control (1m):

- For every temperature examined, no potato disc added – to ensure the K⁺ detected is from the potato cells.

Risks and precautions(1m)

Risk	Precaution
Scalpel is sharp and may cause injuries	Take extra care when handling with the scalpel
70°C water bath is hot enough cause scalding	Use cloth/ for insulation